
Euler's Tonnetz, Graph Theory and Geometry

A Tutorial Review with Reproducible Python Sheets

Hale Tureli

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Abstract. The Tonnetz (“tone network”) of Euler arranges the twelve pitch classes so that consonant triads become triangles of a lattice. Read as mathematics, it is simultaneously a graph, a group action and a triangulated surface. This report is a self-contained tutorial that develops the subject from first principles for students while collecting the computational and empirical literature for researchers. We (i) build the Tonnetz as a quotient of $\mathbb{Z}^2$ and prove it is a torus; (ii) study the neo-Riemannian operations P , L , R and the dual “chicken-wire” graph; (iii) enumerate its Hamiltonian cycles, confirming 124 directed (62 undirected) cycles and classifying them into eight orbits under transposition and inversion; (iv) reproduce Catanzaro’s topological classification of generalized Tonnetze for $N = 12$; (v) analyse seventh-chord and edge-labelled Tonnetze, where non-orientable surfaces such as the real projective plane appear; and (vi) place four public corpora (Bach, Beethoven, two pop collections) on the Tonnetz, finding that classical progressions are more Tonnetz-compact than their own shuffled chords while pop progressions are not. Every chapter ends with questions and answers, and every number is reproduced by an executable notebook in the accompanying repository.

Contents

How to read this report	iii
1 Introduction: a map of harmony	1
1.1 Why a map?	1
1.2 A short history	1
1.3 What this report adds	2
2 Mathematical preliminaries	4
2.1 Pitch classes and intervals	4
2.2 Transposition, inversion and the dihedral group	4
2.3 Graphs	4
2.4 Simplicial complexes, Euler characteristic and surfaces	5
3 The Tonnetz and why it is a torus	7
3.1 Construction as a lattice quotient	7
3.2 Identifying equal notes: the torus	7
3.3 Reading chords and progressions	8
4 P, L, R and the dual graph	10
4.1 Three parsimonious operations	10
4.2 The dual graph	10
4.3 Faces and famous cycles	10
4.4 The <i>PLR</i> group and its dual	11
4.5 Two metrics on triads	12
5 Hamiltonian cycles through all 24 triads	13
5.1 The question	13
5.2 Enumeration by backtracking	13
5.3 Classification up to symmetry	13
5.4 Do they sound better?	14
6 Generalized Tonnetze and their topology	16
6.1 From one trichord to a whole space	16
6.2 All twelve cases for $N = 12$, computed	16
6.3 A parity phenomenon: cylinder or Möbius band	16
7 Seventh chords, edge Tonnetze and non-orientable surfaces	19
7.1 From triads to larger chords	19
7.2 Reconstructing the slide: a triangulated projective plane	19
7.3 The pitch-class complex tells a different story	20
7.4 Edge-labelled gluings: a census	20
7.5 Non-orientable spaces elsewhere in music theory	21
8 The Tonnetz meets data	23
8.1 Public datasets	23
8.2 A reproducible pipeline	23
8.3 Results	23
8.4 From audio: the tonal centroid	25

9	Software ecosystem and open problems	27
9.1	Tools	27
9.2	Open problems	27
9.3	Conclusion	28
A	Repository guide and reproduction	29
A.1	Layout	29
A.2	Reproducing everything	29
A.3	Map from chapters to sheets	29
A.4	Data licences	29
	Acknowledgements	30

How to read this report

The report is written for two audiences at once. *Students* (final-year school, first-year university, or musicians curious about mathematics) should read Chapters 1–5 in order, running the matching Python sheet after each chapter. *Researchers* can skim Chapters 2–4 for notation and go straight to the computational results in Chapters 5–8 and the open problems in Chapter 9.

Companion repository. All code lives in a single repository with four parts: `tonnetzkit/` (a small Python package), `book/` (ten Jupyter notebooks forming a Jupyter Book of “Python sheets”), `scripts/` (figure generation, data download and the corpus study), and `tests/` (regression tests pinning every numerical claim in the text). Boxes like the one below point to the relevant sheet.

Python sheet

`book/00_welcome.ipynb` gives a five-minute tour: it builds the 24 triads, applies P, L, R , counts Hamiltonian cycles and plays a chord progression, all with no external data.

Conventions. Pitch classes are integers mod 12 with $C = 0, C\sharp = 1, \dots, B = 11$. We write major triads by their root ($C, F\sharp$) and minor triads with a lower-case “m” (A_m). Sharps are used throughout, as in the original slides; in 12-tone equal temperament $A\sharp$ and $B\flat$ name the same pitch class.

Relation to the original presentation. The report expands every slide of the DREAMS 2025 presentation. Where a slide simplified or mis-stated something, an orange box explains the correction; these are offered in the spirit of peer review of our own work, not as criticism of a student presentation.

Introduction: a map of harmony

1.1 Why a map?

Musicians have always used spatial language for harmony: chords are “close” or “distant”, a modulation “travels” to a “remote” key, a progression “returns home”. The Tonnetz makes this language literal. It is a diagram in which every point is a note and every small triangle is a chord, arranged so that geometric nearness tracks a musical notion of nearness: two triangles that share an edge are two chords that share two notes and can be connected by moving a single voice a short distance.

Once the picture is drawn, three branches of mathematics appear at once. As a *graph* it invites questions about paths and cycles: can we walk through all 24 major and minor chords, each exactly once, moving only between neighbours? As a *group action* it explains why transposing a progression preserves its shape. As a *surface* it has a topology: when the diagram is wrapped up so that each note appears only once, it becomes a torus, and variants built from other chords become cylinders, spheres, Möbius bands or even the real projective plane. The goal of this report is to develop all three views carefully, to compute everything that can be computed, and finally to test the map against real music.

1.2 A short history

Euler (1739). Leonhard Euler’s *Tentamen novae theoriae musicae* [25] contains a small lattice of note names in which horizontal neighbours differ by a perfect fifth and vertical neighbours by a major third. For Euler, working in just intonation, the lattice was a way to organise frequency ratios built from the primes 3 and 5; it was not yet a theory of chord progressions.

Nineteenth-century dualism. Arthur von Oettingen [42] and Hugo Riemann [39] revived and extended the lattice (the German *Tonnetz* is Riemann’s usage). Their harmonic “dualism” treated minor triads as mirror images of major ones, an idea that survives in the inversive symmetry of Chapter 2.

Transformational and neo-Riemannian theory. David Lewin’s *Generalized Musical Intervals and Transformations* [34] recast music theory in the language of groups acting on sets. Richard Cohn then isolated three “parsimonious” transformations, P (parallel), L (leading-tone exchange) and R (relative), that move a triad to another triad by shifting one voice by a semitone or a whole tone [16–18]. The Tonnetz became the natural stage for these operations, and a large body of analysis of chromatic nineteenth-century music followed [19], extended later to popular music [12].

Geometry and topology. From the late 1990s mathematicians and theorists studied the Tonnetz as a space. Douthett and Steinbach introduced the “chicken-wire torus”, the dual graph on triads [24]; Crans, Fiore and Satyendra showed that the PLR group is dihedral of order 24 and is the dual of the transposition/inversion group [20]. Tymoczko connected chord spaces to orbifolds and voice leading [9, 40, 41], and Catanzaro classified the topology of every “generalized Tonnetz” built from a single trichord type in any equal temperament [13]. Bigo and collaborators represented chord collections as simplicial complexes and built the HexaChord software [5, 6].

Yust [44], Jevtić and Živaljević [31] and, most recently, Boland and Hughston [7, 8] continue to generalise the construction, the latter linking the Tonnetz to classical point–line configurations.

Computation and data. In parallel, music information retrieval adopted the Tonnetz as a feature: the tonal centroid of Harte, Sandler and Gasser [29] is implemented in the `librosa` library [36] and used in chord recognition, and the Tonnetz has been used as an input representation for neural networks [14] and for genre classification from Tonnetz trajectories [33]. Large open chord corpora, notably ChoCo [22] and Chordonomicon [32], now make it possible to test Tonnetz-based hypotheses at scale; Xu, Hall and Rohrmeier’s recent study of tonal coherence on the Tonnetz across 2,800 pieces [43] is an example.

1.3 What this report adds

Beyond a unified tutorial exposition, the report contributes a set of verified, reproducible computations:

1. an independent enumeration of Hamiltonian cycles in the dual graph (62 undirected, 124 directed, as reported by Albini and Antonini [1]) together with their classification into eight symmetry orbits and their voice-leading costs (Chapter 5);
2. a machine-checked table of all twelve generalized Tonnetze in 12-tone equal temperament, including the parity phenomenon that turns the chromatic strip into a Möbius band for odd N (Chapter 6);
3. a reconstruction of the presentation’s whole-tone “edge Tonnetz” as a triangulated real projective plane, contrasted with the actual pitch-class complex and with a Monte-Carlo census of edge-labelled gluings (Chapter 7);
4. a four-corpus study with a shuffling null model (Chapter 8).

The one-sentence summary

The Tonnetz is a triangulated torus whose vertices are the twelve notes and whose 24 triangles are the major and minor triads; its dual is a 3-regular graph on those triads in which each edge is one of Cohn’s single-voice moves P , L or R .

Questions and answers

- Q.** In one sentence each, what are the three mathematical “views” of the Tonnetz used in this report?
- A.** As a graph (notes or chords joined by musical relations, studied through paths and cycles); as a group action (transposition, inversion and the P , L , R operations acting on chords); and as a surface (a triangulated torus whose topology can be computed).
- Q.** Why was Euler’s original lattice not yet a torus?
- A.** Euler worked in just intonation, where no two lattice points have exactly the same frequency ratio, so the lattice is an infinite plane. The torus appears only after identifying notes that are equal in 12-tone equal temperament (Chapter 3).
- Q.** Which three operations did Cohn call parsimonious, and what makes them parsimonious?
- A.** P , L and R . Each keeps two notes of a triad fixed and moves the third by a semitone (P , L) or a whole tone (R), so the voice leading is as small as possible between a major and a minor triad.

- Q.** Name one audio-processing tool that uses the Tonnetz.
- A.** The tonal-centroid (“tonnetz”) feature of Harte, Sandler and Gasser, available as `librosa.feature.tonnetz`; it projects a chroma vector onto three circles (fifths, minor thirds, major thirds).
- Q.** What is the practical difference between a tutorial review and a research article?
- A.** A tutorial review re-derives known results in a teachable order and points to the literature; this one also adds reproducible code and a few new computations, but its main purpose is to make the field accessible.

Mathematical preliminaries

This chapter collects the small amount of algebra, graph theory and topology used later. Readers comfortable with modular arithmetic, graphs and the Euler characteristic can skip to Chapter 3.

2.1 Pitch classes and intervals

In 12-tone equal temperament (12-TET) the octave is divided into twelve equal semitones. Two identifications turn pitches into *pitch classes*: *octave equivalence* (C4 and C5 are the same class) and *enharmonic equivalence* (F# and Gb are the same key on the piano). The set of pitch classes is then the cyclic group $\mathbb{Z}_{12} = \{0, 1, \dots, 11\}$ with addition mod 12, where C = 0.

An *interval* from x to y is $y - x \bmod 12$. The musically important ones here are the minor third (3), major third (4) and perfect fifth (7). Note that $3 + 4 = 7$ and $3 + 4 + 5 = 12$: a stack of a minor third, a major third and a perfect fourth (5) closes the octave. This numerical coincidence is the reason the Tonnetz exists.

Definition 2.1. A *trichord* is a 3-element subset of $\mathbb{Z}_{12}$. The *major triad* on root r is $\{r, r + 4, r + 7\}$ and the *minor triad* on root r is $\{r, r + 3, r + 7\}$.

Going around the pitch-class circle, the three notes of a trichord cut it into three gaps (n_1, n_2, n_3) with $n_1 + n_2 + n_3 = 12$. The major triad has gaps (4, 3, 5) and the minor triad (3, 4, 5): the same numbers in the opposite cyclic order. This *interval partition* is the key parameter of Catanzaro's generalized Tonnetz (Chapter 6).

2.2 Transposition, inversion and the dihedral group

The *transposition* $T_n(x) = x + n$ and the *inversion* $I_n(x) = n - x$ are bijections of $\mathbb{Z}_{12}$. They satisfy

$$T_m T_n = T_{m+n}, \quad I_m I_n = T_{m-n}, \quad T_m I_n = I_{m+n}, \quad I_n T_m = I_{n-m}.$$

The 24 maps $\{T_n, I_n\}$ form a group, the dihedral group D_{12} of order 24 (the symmetries of a regular 12-gon: rotations are transpositions, reflections are inversions). D_{12} acts on trichords; transposition preserves major and minor, while every inversion swaps them: for example $I_0(\{0, 4, 7\}) = \{0, 8, 5\}$, which is F minor. Hence D_{12} acts *simply transitively* on the 24 consonant triads: for any two triads there is exactly one T_n or I_n carrying one to the other.

Remark 2.2. Under the action of D_{12} the 220 trichords of $\mathbb{Z}_{12}$ fall into 12 orbits (the *trichordal set classes*); under transpositions alone they fall into 19 orbits. Section 6.1 computes the topology of the Tonnetz associated with each of the 12 set classes.

2.3 Graphs

A *graph* $G = (V, E)$ is a set of vertices with a set of edges, each joining two vertices. The *degree* of a vertex is its number of edges; a graph in which every vertex has degree 3 is *cubic* (3-regular). A graph is *bipartite* if its vertices split into two parts with all edges running between the parts. The *distance* between two vertices is the number of edges on a shortest path, and the *diameter* is the largest distance.

Definition 2.3. A *Hamiltonian cycle* in G is a closed walk that visits every vertex exactly once and returns to its start. A graph that has one is *Hamiltonian*.

Unlike its cousin the Eulerian circuit (traverse every *edge* once, decided in linear time by checking vertex degrees), deciding whether a graph is Hamiltonian is NP-complete in general. For the 24-vertex graphs of this report, exhaustive backtracking is instantaneous (Chapter 5).

If a graph is drawn on a surface without crossings, its *dual* has one vertex per face and joins two faces that share an edge. The dual of the triangulated Tonnetz is the graph of triads connected by P , L and R .

2.4 Simplicial complexes, Euler characteristic and surfaces

Definition 2.4. An (abstract) *simplicial complex* is a set V of vertices together with a family of finite subsets (simplices) closed under taking subsets. A 2-dimensional complex has vertices, edges (2-sets) and triangles (3-sets). Its *f-vector* is (V, E, F) and its *Euler characteristic* is $\chi = V - E + F$.

A *chord complex* has pitch classes as vertices and a chosen family of trichords as triangles [6]. A Δ -*complex* is the slightly more general object obtained by gluing triangles along edges, where two triangles may share more than one edge; the Euler characteristic is defined the same way.

Homology in one paragraph. Orient each edge and triangle. The boundary maps ∂_1 (edge $\mapsto$ end $-$ start) and ∂_2 (triangle $\mapsto$ signed sum of its three edges) are matrices; the *Betti numbers* over a field k are

$$b_0 = V - \text{rk } \partial_1, \quad b_1 = E - \text{rk } \partial_1 - \text{rk } \partial_2, \quad b_2 = F - \text{rk } \partial_2,$$

and satisfy $\chi = b_0 - b_1 + b_2$. Informally b_0 counts components, b_1 independent loops and b_2 enclosed cavities. Computing them over both $\mathbb{Q}$ and $\mathbb{Z}_2$ detects non-orientability: a closed non-orientable surface has $b_2 = 0$ over $\mathbb{Q}$ but $b_2 = 1$ over $\mathbb{Z}_2$.

Theorem 2.5 (Classification of closed surfaces). *Every connected closed surface is homeomorphic to exactly one of: the sphere S^2 , a connected sum of $g \geq 1$ tori (orientable, $\chi = 2 - 2g$), or a connected sum of $k \geq 1$ real projective planes (non-orientable, $\chi = 2 - k$). It is determined by its Euler characteristic and its orientability.*

This is exactly the theorem quoted on the presentation's slide "Now how does the surface look?". Orientable surfaces always have *even* χ . Consequently a closed surface with $\chi = 1$ must be non-orientable, with $k = 1$: it is the real projective plane $\mathbb{RP}^2$ (Chapter 7). Surfaces with boundary satisfy analogous rules: the cylinder (annulus) and the Möbius band both have $\chi = 0$ and are distinguished by orientability.

Testing orientability by computer. A triangulated surface is orientable if one can choose a cyclic orientation for every triangle so that each shared edge is traversed in opposite directions by its two triangles. A breadth-first search that propagates the orientation from triangle to triangle either succeeds or finds a contradiction; this is `SimplicialComplex.is_orientable()` in `tonnetzkit`.

Python sheet

`book/01_pitch_classes.ipynb`: pitch-class arithmetic, the 24 permutations $\{T_n, I_n\}$ of the triads, and the 19 transposition classes and 12 set classes of trichords.

Questions and answers

Q. Which two equivalences turn pitches into the 12 pitch classes?

A. Octave equivalence and enharmonic equivalence (the latter is exact only in equal

temperament).

- Q.** Verify that I_0 maps C major to F minor.
A. $I_0(x) = -x$: $\{0, 4, 7\} \mapsto \{0, 8, 5\} = \{F, A\flat, C\}$, the F minor triad.
- Q.** Why does D_{12} act simply transitively on the 24 consonant triads?
A. There are 24 group elements and 24 triads; transpositions move C major to every major triad, inversions move it to every minor triad, and no nontrivial element fixes C major (its three gaps are all different, so it has no symmetry). An orbit of size 24 with trivial stabiliser means simple transitivity.
- Q.** What is the difference between a Hamiltonian cycle and an Eulerian circuit?
A. A Hamiltonian cycle visits every vertex once; an Eulerian circuit traverses every edge once. The latter exists in a connected graph iff all degrees are even; the former has no such simple criterion.
- Q.** A closed surface has $\chi = -2$. What can it be?
A. Either the orientable surface of genus 2 ($2 - 2g = -2$) or the non-orientable connected sum of four projective planes ($2 - k = -2$). Orientability decides which.
- Q.** How can homology over two different fields reveal non-orientability?
A. For a closed connected surface, $b_2 = 1$ over $\mathbb{Z}_2$ always, but $b_2 = 1$ over $\mathbb{Q}$ only if it is orientable. A mismatch ($b_2 = 0$ over $\mathbb{Q}$) therefore certifies non-orientability.

The Tonnetz and why it is a torus

3.1 Construction as a lattice quotient

Take the integer lattice $\mathbb{Z}^2$ and the group homomorphism

$$\varphi : \mathbb{Z}^2 \rightarrow \mathbb{Z}_{12}, \quad \varphi(i, j) = 7i + 4j \pmod{12}.$$

Place lattice point (i, j) in the plane at $i\mathbf{e}_1 + j\mathbf{e}_2$ with $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, and label it with the note $\varphi(i, j)$. Then a step along $\mathbf{e}_1$ adds a perfect fifth, a step along $\mathbf{e}_2$ adds a major third, and a step along $\mathbf{e}_1 - \mathbf{e}_2$ adds $7 - 4 = 3$, a minor third. Joining each point to its six neighbours triangulates the plane (Figure 3.1); this is the layout of the TonnetzViz application [15] used on the original slides.

Proposition 3.1. *The upward triangle with lower-left corner $x = \varphi(i, j)$ carries the notes $\{x, x + 7, x + 4\}$, the major triad on x ; the downward triangle on $(i+1, j), (i, j+1), (i+1, j+1)$ carries $\{x + 7, x + 4, x + 11\}$, the minor triad on $x + 4$. Every major and minor triad occurs.*

Proof. Direct substitution into φ . Since φ is surjective (already $\varphi(i, 0) = 7i$ runs through all residues because $\gcd(7, 12) = 1$), every root x occurs, so every major and every minor triad occurs. $\square$

3.2 Identifying equal notes: the torus

Each note appears infinitely often in the plane. To obtain one copy of each, identify lattice points with equal labels, i.e. form the quotient $\mathbb{Z}^2 / \ker \varphi$. The kernel is a sublattice of index 12, and one checks that

$$\ker \varphi = \{(i, j) : 7i + 4j \equiv 0\} = \mathbb{Z}(4, -1) \oplus \mathbb{Z}(0, 3).$$

Musically, $(0, 3)$ says that three major thirds make an octave ($C \rightarrow E \rightarrow G \sharp \rightarrow C$), and $(4, -1)$ that four fifths up and a major third down return to the start ($C \rightarrow G \rightarrow D \rightarrow A \rightarrow E \rightarrow C$). The determinant of the basis is $4 \cdot 3 - (-1) \cdot 0 = 12$, confirming the index.

Theorem 3.2. *The quotient of the triangulated plane by $\ker \varphi$ is a torus triangulated with $V = 12$ vertices, $E = 36$ edges and $F = 24$ triangles; $\chi = 0$. Its triangles are exactly the 24 major and minor triads.*

Proof. A quotient of $\mathbb{R}^2$ by a rank-2 lattice of translations is a torus (glue opposite sides of the fundamental parallelogram, Figure 3.2). The lattice triangulation is invariant under the translations, so it descends. Each vertex has six neighbours (at $\pm 3, \pm 4, \pm 7$, which are six distinct residues), so $E = 12 \cdot 6/2 = 36$; each vertex lies in six triangles, so $F = 12 \cdot 6/3 = 24$. By Proposition 3.1 the 24 triangles are the 24 triads, each once. Finally $12 - 36 + 24 = 0$. $\square$

A computer check of Theorem 3.2 is one line of `tonnetzkit`:

```
K = generalized_tonnetz(3, 4, 5) # all T/I images of {0, 3, 7}
K.summary() # V=12, E=36, F=24, chi=0, betti (1,2,1), orientable -> torus
```

The Betti numbers $(1, 2, 1)$ are those of the torus: one component, two independent loops (the a - and b -circles), one enclosed 2-cycle.

The neo-Riemannian Tonnetz: C major (red) and its P, L, R neighbours (blue)

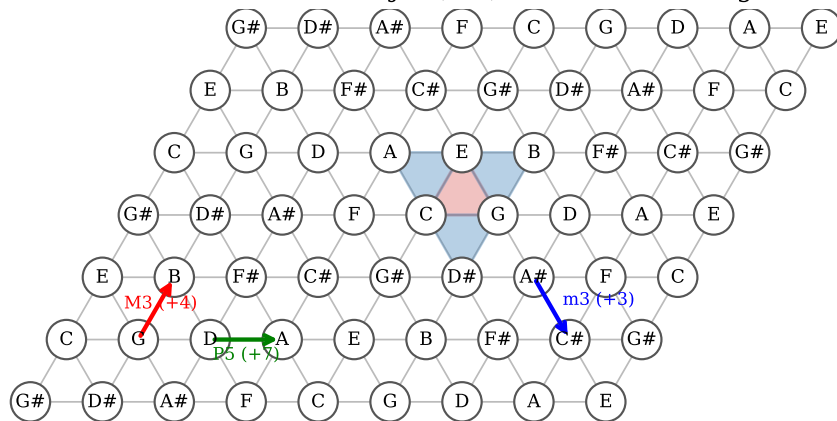


Figure 3.1: The Tonnetz as a labelled triangular lattice. Upward triangles $\{x, x + 4, x + 7\}$ are major triads; downward triangles are minor. C major (red) shares an edge with each of Cm, Em and Am (blue): these are its *P*, *L* and *R* neighbours (Chapter 4).

A note on the original slides

The slide “Hamiltonian cycles on a torus” says the Tonnetz becomes a torus “when you do octaves”. Octave equivalence alone is not enough: in just intonation, as Harte, Sandler and Gasser point out, the Tonnetz is an infinite plane even when octaves are ignored [29]. What closes the plane into a torus is *pitch-class* equivalence in 12-TET, which additionally equates enharmonic spellings. In particular three pure major thirds $(5/4)^3 = 125/64$ fall short of an octave; only in equal temperament do they close exactly.

3.3 Reading chords and progressions

Because transposition T_n acts on the lattice by translation, any chord or melody has a *shape* on the Tonnetz that is unchanged by transposition: all major triads are upward triangles and all minor triads downward. Inversion I_n acts by a point reflection, which is why it swaps the two triangle orientations. A progression becomes a path of triangles; progressions whose consecutive chords share an edge move by one voice, and progressions that “jump” across the diagram involve larger voice leading.

Other layouts. Replacing φ by $7i + 3j$ (fifths and minor thirds) gives a mirror-image drawing of the same complex, because the three generating intervals $\{3, 4, 5\}$ are unchanged. Genuinely different Tonnetze arise only from different trichords (Chapter 6) or from higher-dimensional lattices, for instance Gollin’s three-dimensional Tonnetz for seventh chords [26].

Python sheet

book/02_tonnetz_torus.ipynb: print the lattice, find the kernel vectors by brute force, and verify Theorem 3.2.

Questions and answers

- Q.** Starting from C, which note is reached by the lattice step $(2, 1)$?
A. $\varphi(2, 1) = 14 + 4 = 18 \equiv 6$, i.e. F♯: two fifths ($C \rightarrow G \rightarrow D$) and a major third ($D \rightarrow F^\sharp$).

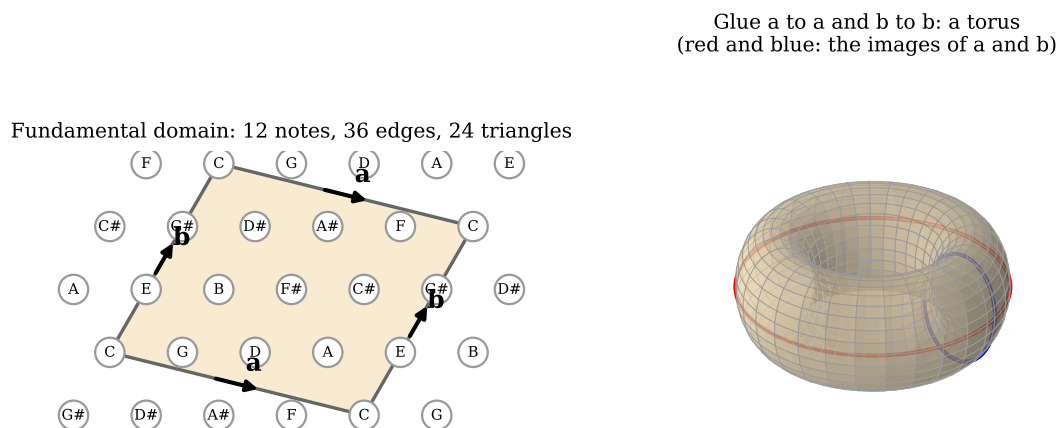


Figure 3.2: Left: a fundamental domain spanned by $(4, -1)$ and $(0, 3)$; all four corners are C. Gluing edge a to a and b to b (right) yields the torus.

- Q.** Explain in musical terms why $(0, 3) \in \ker \varphi$.
A. Three consecutive major thirds, $C-E-G\sharp-C$, return to the starting pitch class (in 12-TET).
- Q.** Why does each vertex of the Tonnetz have exactly six neighbours?
A. Its neighbours are at intervals $\pm 3, \pm 4, \pm 7$, i.e. $\bar{3}, 9, 4, 8, \bar{7}, 5$: six distinct pitch classes.
- Q.** Compute χ from the f-vector and name the surface.
A. $12 - 36 + 24 = 0$; the complex is orientable, so by Theorem 2.5 it is the torus.
- Q.** A student claims the Tonnetz is a sphere because “it has no edges”. What is wrong?
A. Having no boundary does not determine the surface. Both the sphere and the torus are closed; the Euler characteristic (0, not 2) and the two independent loops ($b_1 = 2$) show it is a torus.
- Q.** Why is every major triad an upward triangle?
A. Transposition acts by translation and preserves orientation; C major is upward, and every major triad is a transposition of it.

P, L, R and the dual graph

4.1 Three parsimonious operations

Two triangles of the Tonnetz share an edge exactly when the two triads share two notes. For each triad there are three such neighbours, one across each edge, and they define three involutions on the set of 24 triads:

P (parallel) keeps the root and fifth and moves the third by a semitone: $C \leftrightarrow C_m$.

L (leading-tone exchange) keeps the third and fifth of a major triad and lowers the root a semitone: $C \leftrightarrow E_m$ ($C-E-G \rightarrow B-E-G$).

R (relative) keeps the root and third of a major triad and raises the fifth a whole tone: $C \leftrightarrow A_m$ ($C-E-G \rightarrow C-E-A$).

In the lattice picture (Figure 3.1), *P* flips a triangle across its fifth-edge, *L* across its minor-third edge and *R* across its major-third edge. Each is its own inverse ($P^2 = L^2 = R^2 = \text{id}$) and each swaps major and minor. Measured as total semitone motion of the voices, *P* and *L* cost 1 and *R* costs 2; these are the smallest possible voice leadings between a major and a minor triad, which is what Cohn means by *parsimonious* [17].

4.2 The dual graph

Definition 4.1. The *PLR graph* Γ has the 24 consonant triads as vertices and an edge labelled *P*, *L* or *R* between t and $P(t)$, $L(t)$, $R(t)$.

Γ is the dual of the triangulated Tonnetz: vertices of Γ are triangles of the Tonnetz and edges of Γ cross edges of the Tonnetz. Drawn in the plane it is a hexagonal honeycomb (Figure 4.1), and on the torus it is Douthett and Steinbach’s “chicken-wire torus” [24].

Proposition 4.2. Γ is connected, cubic, bipartite (majors versus minors) and has 24 vertices, 36 edges and diameter 5. From any triad, the numbers of triads at distance 1, 2, 3, 4, 5 are 3, 6, 8, 5, 1.

Proof sketch. Cubic and bipartite follow from the definitions (*P*, *L*, *R* are distinct and each changes the mode). Edges: $24 \cdot 3/2 = 36$. The distance profile is the same from every vertex because D_{12} acts transitively and commutes with *P*, *L*, *R* (below); it is checked by breadth-first search in `book/03_plr_dual.ipynb`. The unique triad at distance 5 from *C* major is *B* minor. $\square$

4.3 Faces and famous cycles

Three families of short cycles in Γ are central to neo-Riemannian analysis.

Hexagons: the triads around a note. Alternately applying *P*, *L*, *R* six times returns to the start. From *C*, $(PLR)^2$ visits *C*, *C_m*, *A_b*, *F_m*, *E*, *A_m*: precisely the six triads that contain the note *C*. These hexagons are the faces of Γ on the torus; there are 12 of them, one per pitch class, and Γ has $V - E + F = 24 - 36 + 12 = 0$, as a graph embedded in the torus must.

Hexatonic cycles. $(PL)^3$ from *C* gives *C*, *C_m*, *A_b*, *A_{b_m}*, *E*, *E_m*: the six triads contained in the hexatonic scale $\{0, 3, 4, 7, 8, 11\}$. There are four such cycles, Cohn’s *hexatonic systems* [16], central to his account of late-Romantic “maximally smooth” progressions.

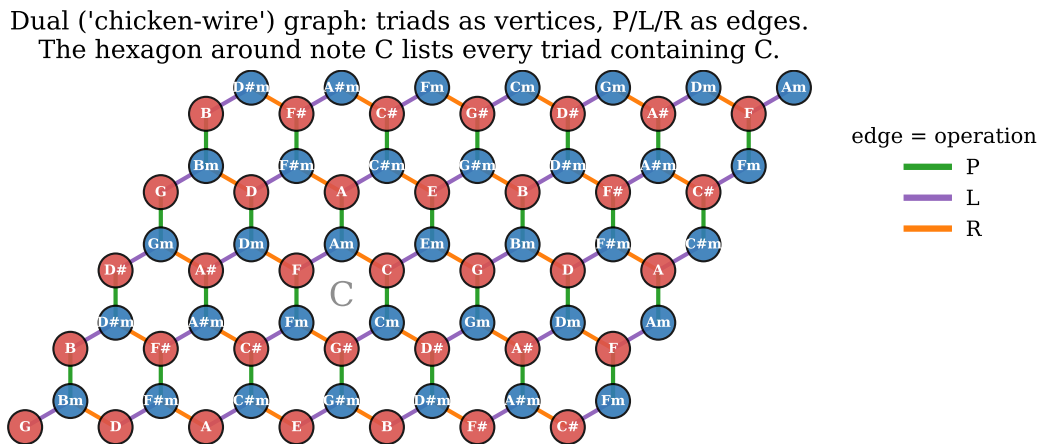


Figure 4.1: A patch of the dual graph. Red vertices are major and blue are minor triads; edge colour gives the operation. Every hexagonal face corresponds to one note of the Tonnetz and lists the six triads containing it.

Octatonic cycles. $(PR)^4$ gives eight triads (C, Cm, Eb, Ebm, F#, F#m, A, Am) in an octatonic scale; there are three such cycles.

The LR chain. $(LR)^{12}$ visits all 24 triads, alternating major and minor while the major roots descend (or ascend) by fifths: C, Em, G, Bm, D, ... Long stretches of this chain occur in the repertoire; a celebrated example is a passage in the scherzo of Beethoven's Ninth Symphony discussed by Cohn [17]. We meet this cycle again as the most symmetric Hamiltonian cycle in Chapter 5.

A note on the original slides

The slide "Hamiltonian cycles (again)" states that "going around a hexagon" gives $C \rightarrow Am \rightarrow F \rightarrow Dm \rightarrow G \rightarrow Em \rightarrow C$. Checking each step against Γ shows that five steps are edges (R, L, R and then R, L) but $Dm \rightarrow G$ is not: $D-F-A$ and $G-B-D$ share only one note (PLR distance 3). The progression is a diatonic chain of falling thirds, not a face of the dual graph. The hexagon through C and Am is $C-Cm-Ab-Fm-F-Am$ (the triads containing C), or, around the note E, $C-Em-E-C\sharp m-A-Am$.

4.4 The PLR group and its dual

Let $\mathcal{G} = \langle P, L, R \rangle$ be the group of permutations of the 24 triads generated by the three involutions. A direct computation (book/03_plr_dual.ipynb) shows $|\mathcal{G}| = 24$; in fact $\mathcal{G}$ is already generated by L and R, since RL has order 12, and $\mathcal{G} \cong D_{12}$. The deeper structural fact is:

Theorem 4.3 (Crans–Fiore–Satyendra [20]). *The PLR group and the T/I group are both dihedral of order 24, each acts simply transitively on the consonant triads, and each is the centraliser of the other in the symmetric group on the 24 triads.*

Informally: P, L and R are defined by *internal* features of a chord ("keep root and fifth") rather than by absolute pitch, so they commute with transposition and inversion. For example $P(T_5 t) = T_5 P(t)$ for every triad t ; the notebook verifies this for all 24 triads. Transformational theorists call this relationship *duality* (Lewin's "simply transitive group action" [34]); it explains why a neo-Riemannian analysis can be transposed freely.

4.5 Two metrics on triads

Distances in Γ count single-voice moves; the *voice-leading distance* instead minimises the total number of semitones moved, over all ways of pairing the notes. They agree on edges only up to the factor that R costs 2, and diverge more for larger moves: $C \rightarrow G$ (a fifth up) is two steps in Γ (L then R) with voice-leading cost 3, while $F \rightarrow G$ (a whole step, as in $IV \rightarrow V$) is four steps in Γ with voice-leading cost 6. This difference matters for the corpus study of Chapter 8.

Python sheet

book/03_plr_dual.ipynb: build Γ , verify Proposition 4.2, list the hexagonal, hexatonic and octatonic cycles, test the slide's progression edge by edge, and generate $\langle P, L, R \rangle$.

Questions and answers

- Q.** Compute P , L and R of A minor.
A. $P(\text{Am}) = A (C \rightarrow C\sharp)$, $L(\text{Am}) = F (E \rightarrow F)$, $R(\text{Am}) = C (A \rightarrow G)$.
- Q.** Why is Γ bipartite?
A. Each of P , L , R changes the mode, so every edge joins a major triad to a minor triad.
- Q.** How many hexagonal faces does Γ have on the torus, and why?
A. Twelve: each face is the set of triads containing a fixed note, and there are 12 notes. Check: $\chi = 24 - 36 + 12 = 0$.
- Q.** Which triads lie on the hexatonic cycle through D major?
A. $(PL)^3$ from D: D, Dm, Bb, Bbm, Fb, Fbm.
- Q.** Why do P, L, R commute with transposition?
A. They are defined relative to a chord's own root, third and fifth, and transposition moves all three together; so transposing first and then applying P gives the same result as applying P then transposing.
- Q.** Give a progression that is short in voice-leading terms but long in Γ .
A. $F \rightarrow G$: every voice moves a whole step (cost 6, small as a gesture and idiomatic as $IV \rightarrow V$) but it takes four $P/L/R$ steps, e.g. $F \xrightarrow{R} Dm \xrightarrow{L} Bb \xrightarrow{R} Gm \xrightarrow{P} G$.

Hamiltonian cycles through all 24 triads

5.1 The question

Is there a chord progression that visits each of the 24 major and minor triads exactly once, moving only by P , L or R , and returns to where it started? In graph language: is Γ Hamiltonian, and if so, how many Hamiltonian cycles does it have? The question was posed and answered by Albin and Antonini [1], who enumerated and classified the cycles and discussed their compositional use; Albin and Bernardi later developed Hamiltonian graphs as harmonic tools more broadly [2]. The presentation slide asks the natural follow-up: *do they sound better?*

Two counting conventions must be fixed first. A cycle can be traversed in two directions and started at any of its 24 vertices. We call a *directed* cycle a cyclic sequence with a chosen direction, and an *undirected* cycle the underlying set of 24 edges. Each undirected cycle corresponds to exactly two directed ones.

5.2 Enumeration by backtracking

Because every vertex of Γ has degree 3, a depth-first search that extends a path one vertex at a time and backtracks on dead ends explores at most 2^{23} branches; in practice far fewer survive. The implementation in `tonnetzkit.graphs` is:

```
def hamiltonian_cycles(G, start):
    path, seen, out = [start], {start}, []
    def rec(v):
        if len(path) == len(G):
            if G.has_edge(v, start) and path[1] < path[-1]: # keep one direction
                out.append(list(path))
            return
        for w in G.neighbors(v):
            if w not in seen:
                seen.add(w); path.append(w); rec(w); path.pop(); seen.discard(w)
    rec(start); return out
```

Fixing the start vertex removes the 24-fold rotational redundancy; the comparison `path[1] < path[-1]` keeps one of the two directions. On a laptop the search finishes in about 0.1 seconds.

Proposition 5.1. Γ has exactly 62 undirected Hamiltonian cycles, i.e. 124 directed Hamiltonian cycles. Starting from a fixed triad there are 470 directed Hamiltonian paths, of which 124 end at a neighbour of the start and close into cycles.

The count 124 agrees with Albin and Antonini [1], and the slide’s “62 such cycles up to ordering” is the undirected count. Both are pinned by the regression test `test_hamiltonian_counts`.

5.3 Classification up to symmetry

Since T_n and I_n commute with P, L, R they map Hamiltonian cycles to Hamiltonian cycles, so the 62 cycles fall into orbits under D_{12} . Each orbit has size $24/|\text{Stab}|$, where the stabiliser is the subgroup of symmetries mapping the cycle to itself. Independently, one can read the sequence of edge labels around each cycle, a cyclic word of length 24 in the letters P, L, R , and consider it

Orbit	#P	#L	#R	VL (semitones)	Cyclic word (up to rotation/reversal)
1	0	12	12	36	$(LR)^{12}$
2	6	6	12	36	$(LRPR)^6$
3	8	12	4	28	$(LPLPLR)^4$
4	9	3	12	36	$(LRPRPRPR)^3$
4	6	6	12	36	$(LRLRPRPR)^3$
12	11	9	4	28	LPLPLRPLPLPRPLPLPRPLPLPR
12	7	9	8	32	LPLRLPRLRPRLRPRLRPRLRPRLR
24	7	9	8	32	LPLRLPRLRLPLRPLPRPLRPRLR
Total: $1 + 2 + 3 + 4 + 4 + 12 + 12 + 24 = 62$ undirected cycles in 8 orbits.					

Table 5.1: The eight classes of Hamiltonian cycles of Γ . Voice-leading cost counts 1 semitone for each P and L and 2 for each R .

up to rotation and reversal. Because symmetries preserve labels, cycles in the same orbit have the same word; our computation shows the converse also holds here.

Table 5.1 and Figure 5.1 summarise the result. Several observations are worth drawing out.

1. The $(LR)^{12}$ cycle is the unique cycle fixed by every symmetry. It is the “circle of fifths with relatives”: $C, Em, G, Bm, D, F\sharp m, \dots$
2. Every cycle uses R at least four times and at most twelve. Because R is the only operation that moves a voice by a whole tone, the total voice leading ranges from 28 to 36 semitones: the classes of orbit size 3 and 12 with only four R moves are the *smoothest* complete tours of the triads.
3. Two pairs of classes share the same letter counts (6/6/12 and 7/9/8) but differ in word order, so the letter counts alone do not classify cycles.

5.4 Do they sound better?

The presentation asks whether Hamiltonian cycles sound better than other progressions. The mathematics cannot answer this, and to our knowledge no controlled perceptual study has addressed it. What can be said is structural: a Hamiltonian cycle uses maximal harmonic variety (every triad once) with minimal voice leading at every step, a combination composers have exploited deliberately: Cannas and Andreatta point to pieces by Giovanni Albin and Moreno Andreatta built on such cycles [10]. A proper test would compare listener ratings of Hamiltonian cycles against controls such as random walks on Γ (same step size, but with repeats) and random orderings of the 24 triads (same variety, larger steps), matched for tempo, register and voicing. Chapter 9 returns to this as an open project, and `book/09_research_sandbox.ipynb` generates candidate stimuli.

Python sheet

`book/04_hamiltonian.ipynb`: enumerate the cycles, reproduce Table 5.1, and *listen* to the smoothest cycle and to the $(LR)^{12}$ cycle, synthesised in the notebook.

Questions and answers

- Q.** Reconcile the numbers 62 and 124.
A. 124 counts directed cycles (a direction of travel is chosen); each undirected cycle gives two directed ones, so there are 62 undirected cycles.
- Q.** Why does fixing the start vertex not lose any cycles?

The 62 Hamiltonian cycles of the PLR graph fall into 8 classes under transposition/inversion

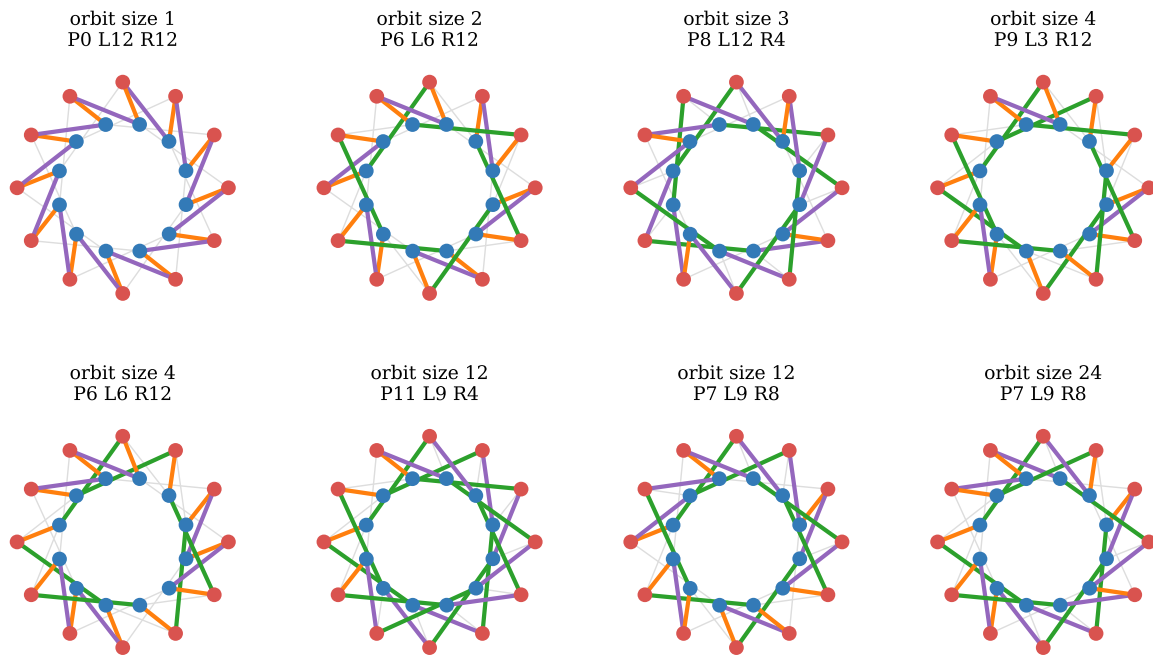


Figure 5.1: One representative of each class. Majors (red) lie on the outer ring ordered by fifths, each relative minor (blue) just inside. Edge colours: P green, L purple, R orange.

A. Every Hamiltonian cycle passes through every vertex, in particular through the chosen start.

Q. What is the orbit size of a cycle whose stabiliser has order 8?

A. $24/8 = 3$ (orbit–stabiliser theorem). This is the third row of Table 5.1.

Q. Explain why the $(LR)^{12}$ cycle is unique among cycles using only L and R .

A. L and R are involutions, so a walk using only them must alternate (repeating a letter would return to the previous triad). The two alternating walks from a given triad are the same cycle in opposite directions, and it closes only after 24 steps because RL has order 12 on major triads.

Q. Which Hamiltonian cycles minimise total voice leading, and what is the minimum?

A. Those with exactly four R moves, the classes of orbit size 3 and 12 (first of the two 12s) in Table 5.1, with 28 semitones.

Q. Design a control condition for the question “do Hamiltonian cycles sound better?”.

A. For example, random walks of 24 steps on Γ (same local smoothness, repeated chords allowed) and random permutations of all 24 triads (same variety, larger steps), rendered identically and rated blind.

Generalized Tonnetze and their topology

6.1 From one trichord to a whole space

The classical Tonnetz is built from a single chord type: all transpositions and inversions of $\{0, 4, 7\}$. Nothing prevents us from starting with any other trichord. Following Catanzaro [13], fix an equal temperament $\mathbb{Z}_N$ and a partition $N = n_1 + n_2 + n_3$ into positive parts, and let

$\mathcal{C}(n_1, n_2, n_3)$ = the chord complex whose triangles are all T_t and I_t images of $\{0, n_1, n_1 + n_2\}$.

Its vertices are the notes that occur, its edges are the note pairs that occur inside some chord, and its triangles are the chords. The classical Tonnetz is $\mathcal{C}(3, 4, 5)$ in $\mathbb{Z}_{12}$. Related constructions include Cohn’s parsimonious trichords [17], Tymoczko’s generalized Tonnetz of voice-leading lattices [41], Bigo et al.’s chord-based complexes [6], and Yust’s Tonnetze and “Zeitnetze” [44]; Jevtić and Živaljević prove that sufficiently generic higher-dimensional analogues triangulate tori T^{k-1} [31].

Catanzaro’s theorem [13] classifies all these spaces for every N : the torus is the most common outcome, especially as N grows; the other possibilities are single 2-simplices, boundaries of tetrahedra, the “harmonic strip” (a cylinder or Möbius band), and, for trichords containing a tritone, a “circle of tetrahedra boundaries” related to Peck’s Klein-bottle Tonnetz.

6.2 All twelve cases for $N = 12$, computed

Up to transposition and inversion there are twelve trichord types in 12-TET. Table 6.1 lists the complex of each, computed with `generalized_tonnetz` and classified by the procedure of Chapter 2: count components, check that every edge lies in at most two triangles, compute χ and orientability, then apply Theorem 2.5.

The table rewards careful reading.

- *Six of the twelve types give tori*, including the classical Tonnetz. For a torus, each vertex must have six distinct neighbours ($\pm n_1, \pm n_2, \pm n_3$ distinct), giving $E = 36$ and $F = 24$.
- *Symmetric chords collapse the space*. The augmented triad is fixed by T_4 , so only four exist and they are disjoint. The diminished triad lives inside a diminished-seventh collection of four notes whose four 3-subsets form the boundary of a tetrahedron, a sphere with $\chi = 2$; three such collections give $\chi = 6$.
- *Repeated gaps give strips*. When two gaps are equal (and no gap is a tritone or a third of the octave), the chord’s neighbours coincide in pairs and the complex is a long strip closing up on itself: a cylinder for $N = 12$.
- *Tritones break the surface*. If a gap is 6, the tritone edge $\{x, x + 6\}$ equals $\{x + 6, x + 12\}$ and is shared by four triangles, so no neighbourhood looks like a disc. These are Catanzaro’s circles of tetrahedra boundaries.

6.3 A parity phenomenon: cylinder or Möbius band

The chromatic cluster $\mathcal{C}(1, 1, N - 2)$ is a strip of triangles $\{x, x + 1, x + 2\}$ with the twisted edges of consecutive triangles glued. Whether the strip closes with or without a half-twist depends on

(n_1, n_2, n_3)	Example chord	V	E	F	χ	Space (Betti numbers over $\mathbb{Q}$)
(1,1,10)	C–C♯–D cluster	12	24	12	0	cylinder (harmonic strip); (1,1,0)
(1,2,9)	C–C♯–D♯	12	36	24	0	torus; (1,2,1)
(1,3,8)	C–C♯–E	12	36	24	0	torus; (1,2,1)
(1,4,7)	C–C♯–F	12	36	24	0	torus; (1,2,1)
(1,5,6)	C–C♯–F♯	12	30	24	6	not a surface (edges in 4 triangles); (1,1,6)
(2,2,8)	C–D–E	12	24	12	0	two cylinders (one per whole-tone scale); (2,2,0)
(2,3,7)	C–D–F	12	36	24	0	torus; (1,2,1)
(2,4,6)	C–D–F♯	12	30	24	6	two components, not surfaces; (2,2,6)
(2,5,5)	C–D–G	12	24	12	0	cylinder; (1,1,0)
(3,3,6)	diminished C–E♭–G♭	12	18	12	6	three tetrahedron boundaries (spheres); (3,0,3)
(3,4,5)	major/minor triads	12	36	24	0	torus ; (1,2,1)
(4,4,4)	augmented C–E–G♯	12	12	4	4	four disjoint triangles; (4,0,0)

Table 6.1: Generalized Tonnetze in 12-TET. Betti numbers over $\mathbb{Z}_2$ coincide with those over $\mathbb{Q}$ in every row. Reproduced in book/05_generalized_tonnetze.ipynb; consistent with Catanzaro’s classification.

the parity of N . Computing orientability for $5 \leq N \leq 15$ gives

$$\mathcal{C}(1, 1, N - 2) \cong \begin{cases} \text{cylinder,} & N \text{ even,} \\ \text{Möbius band,} & N \text{ odd.} \end{cases}$$

This is Catanzaro’s harmonic strip “in both its cylinder and Möbius band variants”. The Möbius case is the first non-orientable space in this report; Chapter 7 meets closed non-orientable surfaces.

Beyond 12-TET. In 19-tone equal temperament, looping over all 51 transposition classes of trichords with `generalized_tonnetz(a, b, c, n=19)` gives 42 tori and 9 Möbius bands: since 19 is prime and odd, no chord has transpositional symmetry and all harmonic strips twist. This illustrates the “torus is generic” part of Catanzaro’s theorem.

Remark 6.1 (Why compute homology at all?). Euler characteristic and orientability suffice for closed surfaces, but many musical complexes are not surfaces. Betti numbers still describe them: $\mathcal{C}(1, 5, 6)$ has $b_2 = 6$ over $\mathbb{Q}$, six independent enclosed 2-cycles, one per tritone pair. Topological signatures of this kind underlie the “compliance” measure of Bigo et al. [6] and persistent-homology fingerprints of musical pieces [3].

Python sheet

book/05_generalized_tonnetze.ipynb: regenerate Table 6.1 as a data frame, watch the harmonic strip twist for odd N , and extend the table to 19-TET.

Questions and answers

- Q.** How many trichord types are there in 12-TET up to transposition only, and up to transposition and inversion?
A. 19 and 12. Inversion identifies a type with its mirror image, e.g. major (4,3,5) with minor (3,4,5).
- Q.** Why does $\mathcal{C}(4, 4, 4)$ have only four triangles?
A. The augmented triad is invariant under T_4 and under inversion, so its 24 images coincide in groups of six: $24/6 = 4$ distinct triads, pairwise disjoint.
- Q.** Show that each component of $\mathcal{C}(3, 3, 6)$ is a sphere.

A. A component is a diminished-seventh collection $\{x, x + 3, x + 6, x + 9\}$; every 3-subset is a diminished triad, so the four triangles form a tetrahedron boundary:
 $V - E + F = 4 - 6 + 4 = 2$.

Q. Why is $\mathcal{C}(1, 5, 6)$ not a surface?

A. Its tritone edges each lie in four triangles; around such an edge the space looks like four half-planes meeting along a line, not like a plane.

Q. For which N is the chromatic strip a Möbius band?

A. For odd N ; for even N it is a cylinder.

Q. Name two spaces in Table 6.1 with the same Euler characteristic but different topology.

A. The torus and the cylinder both have $\chi = 0$; the torus is closed with $b_1 = 2$, the cylinder has boundary and $b_1 = 1$. Euler characteristic alone does not classify spaces with boundary or with several components.

Seventh chords, edge Tonnetze and non-orientable surfaces

7.1 From triads to larger chords

Four- and five-note chords do not fit on a triangular surface directly. Two strategies appear in the literature. One raises the dimension: Gollin’s three-dimensional Tonnetz places dominant and half-diminished sevenths as tetrahedra [26], and seventh-chord analogues of the *PLR* group have been studied as groups acting on the classical seventh-chord types [11] and through generalized duals of the Tonnetz [10]; Boland and Hughston build a tone network for “Tristan-genus” chords from a point–line configuration [7]. The other strategy, adopted in the presentation, *reduces* each chord to three “essential” notes so that chords remain triangles.

A note on the original slides

The “Edge Tonnetz” slide reduces $C7 = C-E-G-B\flat$ to $C-E-B\flat$ “because the dominant note (G) does not change the way the chord sounds”. The omitted note is the chord’s *fifth*; “dominant” names a chord function (V), not a chord member. The fifth is the standard note to omit because the root already implies it through the overtone series, while the third and seventh define the chord quality. The slide reduces $Cmaj9 = C-E-G-B-D$ to $C-E-D$ (root, third, ninth). That is a legitimate choice, but it discards the major seventh, which distinguishes *maj9* from the dominant-ninth chord; jazz pianists more often keep third, seventh and ninth ($E-B-D$). `tonnetzkit.essential_notes` implements both (“maj9” and “maj9_jazz”).

With the slide’s reductions, the dominant-seventh shell on root r is $\{r, r+4, r+10\}$ and the major-ninth shell is $\{r, r+2, r+4\}$. Both lie inside the whole-tone scale containing r , which is why the presentation builds a “Tonnetz based on the whole tone scale”. Its hand-drawn diagram labels triangles with chords ($A7, G9, F7, \dots$) and *edges* with notes: in an edge Tonnetz, notes are edges rather than vertices.

7.2 Reconstructing the slide: a triangulated projective plane

The diagram on the slide “Now how does the surface look?” is a hexagonal region of side 2 in the triangular lattice: 24 triangles, 19 vertices and 42 edges before gluing. The thick arrows around its border indicate that each boundary point is glued to the *antipodal* boundary point, the point diametrically opposite. After gluing, the 12 boundary vertices become 6 and the 12 boundary edges become 6, so

$$V = 7 + 6 = 13, \quad E = 30 + 6 = 36, \quad F = 24, \quad \chi = 13 - 36 + 24 = 1,$$

matching the slide’s “ $V - E + F = 1$ ”. Figure 7.1 shows the construction.

Proposition 7.1. *The glued hexagon is a triangulation of the real projective plane $\mathbb{RP}^2$. Its Betti numbers are $(1, 0, 0)$ over $\mathbb{Q}$ and $(1, 1, 1)$ over $\mathbb{Z}_2$, and it admits no coherent orientation.*

Proof. A closed disc with antipodal boundary identification is a standard model of $\mathbb{RP}^2$. Independently: every edge lies in two triangles and every vertex link is a single cycle, so

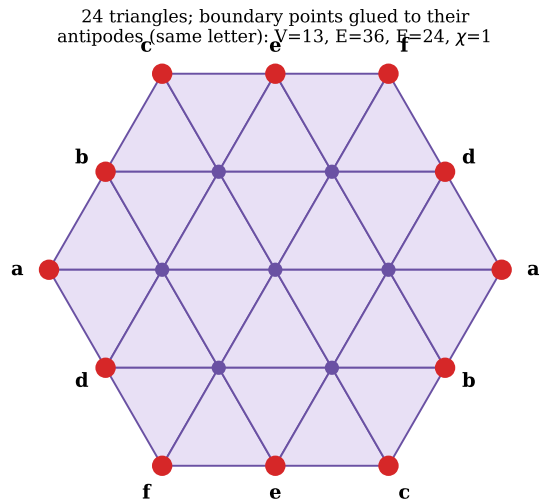


Figure 7.1: The slide’s surface, reconstructed: a hexagonal patch of 24 triangles whose boundary vertices with the same letter (and the boundary edges between them) are identified antipodally. A disc with antipodal boundary points identified is, by definition, the real projective plane.

the complex is a closed surface; the orientation-propagation test fails; and by Theorem 2.5, $\chi = 1$ with non-orientability gives $k = 1$ cross-cap. The Betti numbers are computed by `SimplicialComplex.betti`; the difference between the two fields reflects $H_1(\mathbb{RP}^2; \mathbb{Z}) = \mathbb{Z}_2$. $\square$

The slide’s own reasoning is also correct and worth making explicit: if the surface were orientable, then $2 - 2g = 1$ would force $g = \frac{1}{2}$, which is impossible. The surface therefore *must* be non-orientable, which is exactly why the next slide turns to Möbius strips, Klein bottles and Boy’s surface (an immersion of $\mathbb{RP}^2$ in 3-space).

7.3 The pitch-class complex tells a different story

It is instructive to compare the diagram with the chord complex of Chapter 6, where notes are vertices. Inside one whole-tone scale (6 notes) there are 6 dominant-seventh shells and 6 major-ninth shells:

Triangles	V	E	F	χ	Space
6 dom.-seventh shells $\{r, r+4, r+10\}$	6	15	6	-3	orientable, with boundary
6 major-ninth shells $\{r, r+2, r+4\}$	6	12	6	0	cylinder
both (12 chords)	6	15	12	3	not a surface (see text)

The combined complex has Betti numbers $(1, 0, 2)$ and is not a surface at all. Hence the projective plane of the slide is not the vertex-based chord complex; it is one particular *edge-labelled* gluing, and the choice of gluing matters. The next section quantifies how much.

7.4 Edge-labelled gluings: a census

Definition 7.2. Given a list of chords, each a triangle whose three edges carry its three notes, an *edge Tonnetz* is obtained by gluing all edges in pairs, only ever gluing two edges with the same note label, with one of the two possible orientations for each glued pair.

Gluing polygons edge-to-edge in pairs always produces a closed surface (possibly disconnected); this is the setting of the classification theorem. For the 12 shell chords of the whole-tone scale $\{1, 3, 5, 7, 9, 11\}$ every note belongs to exactly 6 chords, so each note labels 6 edge slots that must be paired into 3 glued edges: 15 perfect matchings times 2^3 orientations per note, i.e. $120^6 \approx 3 \times 10^{12}$ gluings. Every one has $F = 12$ and $E = 18$, hence $\chi = V - 6$, but the number of vertices depends on the gluing.

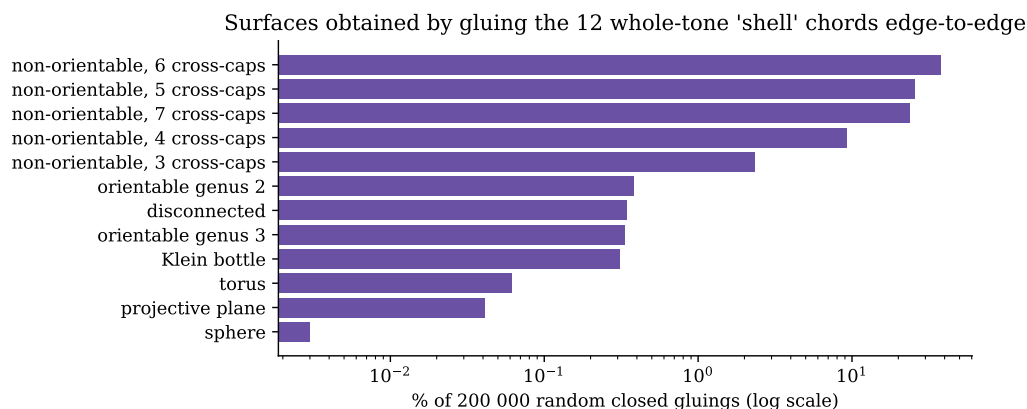


Figure 7.2: Surfaces obtained from 200 000 uniformly random closed gluings of the 12 whole-tone shell chords.

Figure 7.2 shows a Monte-Carlo census of 200 000 random gluings (`sample_edge_tonnetze`). The typical outcome is a non-orientable surface with 5 to 7 cross-caps (together 87%). Orientable surfaces are rare (0.78% in total: genus 2, genus 3, torus, and sphere). The projective plane occurs in only 0.041% of gluings, and the sphere in 0.003%. The surface of the slide is therefore a very special member of a very large family. Two lessons follow.

1. An edge Tonnetz is not determined by its chord vocabulary alone. A musically motivated gluing rule (for instance, glue two chords along a note only if they also share a second note, so that adjacent triangles are parsimoniously related) is needed to make the construction canonical.
2. Finding the gluings with the *largest* Euler characteristic is a combinatorial optimisation problem; we pose exact enumeration up to symmetry as an open problem in Chapter 9.

7.5 Non-orientable spaces elsewhere in music theory

Non-orientability is not an exotic accident. The space of unordered two-note chords is a Möbius band [9, 40]: it is the space of unordered pairs of points on the pitch-class circle, whose single boundary circle consists of the unisons, and in which the tritones form the central circle of the band. Catanzaro's harmonic strip is a Möbius band in odd temperaments (Chapter 6), and his tritone Tonnetze relate to Peck's Klein-bottle Tonnetz [13]. These examples show that the orientable torus of the classical Tonnetz is a special case, not the norm.

Python sheet

`book/06_edge_tonnetz.ipynb`: rebuild the glued hexagon and verify Proposition 7.1, compute the pitch-class complexes above, run the census, and search for a spherical gluing.

Questions and answers

- Q.** In $C7 = C-E-G-B\flat$, which note is usually omitted in a shell voicing, and why?
A. The fifth, G: it adds little to the chord's identity, which is carried by the third (quality) and the seventh (dominant function).
- Q.** Verify the counts $V = 13$, $E = 36$ for the glued hexagon.
A. Before gluing: 19 vertices (7 interior, 12 boundary) and 42 edges (30 interior, 12

boundary). Antipodal gluing halves the boundary: $7 + 6 = 13$ vertices and $30 + 6 = 36$ edges.

- Q. Why must a closed surface with $\chi = 1$ be non-orientable?
A. Orientable closed surfaces have $\chi = 2 - 2g$, always even.
- Q. What are the Betti numbers of $\mathbb{RP}^2$ over $\mathbb{Q}$ and over $\mathbb{Z}_2$?
A. $(1, 0, 0)$ over $\mathbb{Q}$ and $(1, 1, 1)$ over $\mathbb{Z}_2$.
- Q. For the 12-chord edge Tonnetz, how many vertices does a spherical gluing have?
A. $\chi = V - 18 + 12 = 2$ gives $V = 8$.
- Q. Why is the whole-tone pitch-class complex with both chord types not a surface?
A. Each whole-tone edge $\{x, x + 2\}$ lies in two major-ninth shells and one dominant-seventh shell: three triangles, whereas a surface allows at most two.

The Tonnetz meets data

The previous chapters treated the Tonnetz as a mathematical object. This chapter asks how real music moves on it. We survey public datasets and software, describe a reproducible pipeline, and report a small four-corpus study.

8.1 Public datasets

Table 8.1 lists open chord corpora suitable for Tonnetz analysis. They differ in three ways that matter: *representation* (chord labels such as Harte syntax [28], Roman numerals, or full scores), *timing* (aligned to audio or to the score), and *licence*.

8.2 A reproducible pipeline

Parsing. Harte labels such as A:min7/b3 or C:(1,3,5,b7) are parsed into pitch-class sets by `harte_to_pcs`. DCML annotations list chord tones on the *line of fifths* relative to the local tonic, so a tone k has pitch class $\text{tonic} + 7k \bmod 12$; the local tonic is computed from the global key and the Roman-numeral local key (`dcml_chord_pcs`). Bach chorales are “chordified” with `music21` into vertical slices.

Reduction to triads. To place chords on the classical Tonnetz, each chord is reduced to its underlying major or minor triad when it has one (C:7→C, A:min7→Am), following the “majmin” reduction standard in chord-recognition evaluation [38]. Diminished, augmented and suspended chords are dropped; consecutive repeats are merged. For the Beethoven corpus we use the first three DCML chord tones (root, third, fifth), so V⁷ counts as V. For Bach we keep only slices whose pitch-class content is exactly a consonant triad.

Measurement. For each pair of consecutive triads we record their distance in Γ and their voice-leading cost. As a *null model* we shuffle the order of chords within each piece (keeping the piece’s chord vocabulary), merge any repeats created, and recompute the mean distance, averaged over 20 shuffles. A corpus whose progressions are shorter than its own shuffles uses the Tonnetz “locally” beyond what its vocabulary alone implies.

8.3 Results

Table 8.2 and Figure 8.1 support four observations.

1. *All four corpora are far more local than chance.* Distance 2 dominates everywhere. It is the distance of motion by fifth (C→G via Em, i.e. L then R), the backbone of tonal harmony, and of motion by third (C→Am, C→Em are distance 1).
2. *Classical progressions exploit the Tonnetz beyond their vocabulary.* Beethoven’s mean distance (2.39) is well below that of his own shuffled chords (2.58); Bach’s is slightly below (2.33 versus 2.36). The chorale vocabulary is already so diatonic that shuffling it changes little.
3. *Pop progressions do not.* Both pop corpora are *longer* than their shuffles (RWC 2.62 versus 2.47). Almost the whole difference sits at distance 4, which occurs about three times as often in pop as in the classical corpora.

Dataset	Size	Representation	Access
ChoCo [22]	18 integrated collections	JAMS; Harte, Roman, lead-sheet	github.com/smashub/choco , CC BY 4.0 (mostly)
Chordonomicon [32]	>666 000 songs	chord symbols + genre, year, sections	github.com/spyroskantarelis/chordonomicon ; Hugging Face
ABC: Annotated Beethoven Corpus [37]	70 quartet movements	DCML harmony labels (TSV)	github.com/DCMLab/ABC , CC BY-NC-SA 4.0
Distant Listening Corpus [30]	1 326 pieces, 36 composers	DCML standard	github.com/DCMLab/distant_listening_corpus
NYU MARL annotations	100 RWC-Pop + 195 USPop2002 songs	Harte-syntax .lab	github.com/tmc323/Chord-Annotations
music21 corpus [21]	≈400 Bach chorale files	full scores (MusicXML)	ships with pip install music21

Table 8.1: Open corpora for Tonnetz studies. The last four rows are used in this chapter.

Corpus	Pieces	Transitions	Mean d_Γ	Shuffled	$P(d=1)$	$P(d=2)$	$P(d=4)$	Mean VL
Bach chorales	400	13 794	2.33	2.36	0.170	0.447	0.072	3.08
Beethoven (ABC)	70	15 097	2.39	2.58	0.131	0.469	0.075	3.05
RWC Pop	100	11 737	2.62	2.47	0.126	0.367	0.215	3.67
USPop2002 subset	194	21 945	2.49	2.44	0.154	0.415	0.197	3.57
Uniform random	–	–	2.78	–	0.130	0.261	0.217	–

Table 8.2: Consecutive-triad distances on the PLR graph (d_Γ) and voice-leading cost (VL, semitones). “Shuffled” is the mean distance after randomly reordering each piece’s chords. Produced by `scripts/corpus_analysis.py`.

4. *Distance 4 in pop is the whole-step move.* Among distance-4 transitions, major-to-major root motion by a whole step accounts for 77% in USPop (45% up, 32% down) and 67% in RWC. These are $IV \rightarrow V$, $bVII \rightarrow I$ and $I \rightarrow bVII$, idiomatic in rock [4, 23]. In Γ they cost four steps; in voice leading they cost 6 semitones (all three voices move by a tone) against 3 for motion by fifth.

The last point is a caution as much as a finding. The Tonnetz metric encodes a specific theory of closeness (common tones and semitone voice leading). Pop’s parallel whole-step motion is “far” under that theory but is not perceived as a remote relation. Using the Tonnetz as a style descriptor, as in genre classification from Tonnetz trajectories [33], measures distance from that theory, not a universal notion of harmonic distance. Xu, Hall and Rohrmeier reach a related conclusion from pitch distributions: popular music shows higher tonal focus but classical music more structured traversal of the Tonnetz [43], building on the Tonal Diffusion Model [35].

Limitations. The reductions discard information (all seventh chords collapse to triads, and in Bach non-triadic slices are skipped, which can create artificial jumps). The pop annotations are by trained students and the classical ones by experts; annotation conventions differ. The corpora are small by modern standards; running the same script on ChoCo or Chordonomicon would test whether the pattern generalises across genres and decades.

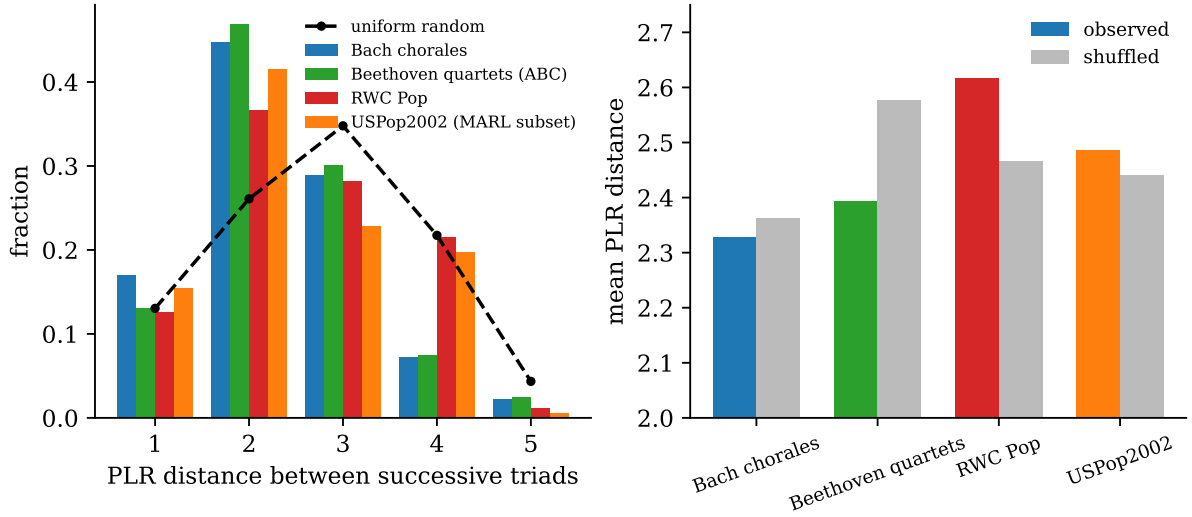


Figure 8.1: Left: distribution of *PLR* distances between successive triads, with the uniform baseline (dashed). Right: observed mean distance versus the within-piece shuffle.

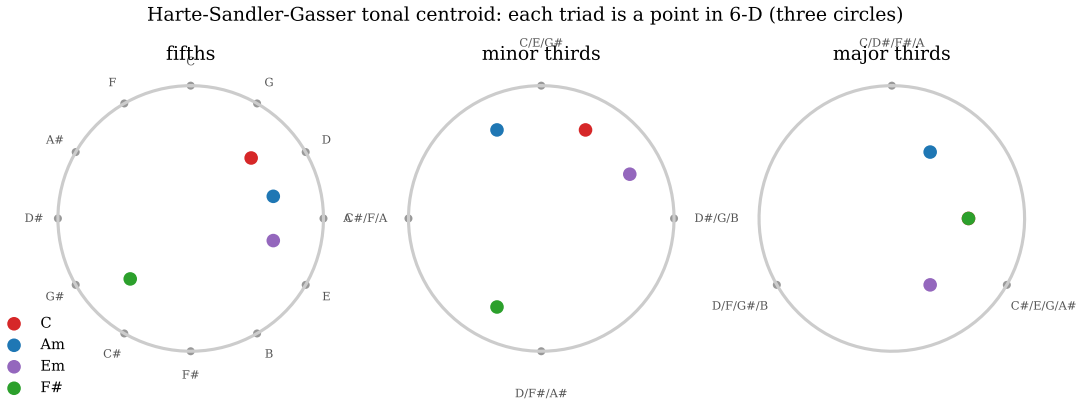


Figure 8.2: Tonal centroids of four triads on the three circles. C and Am (an *R* pair) are close on all three.

8.4 From audio: the tonal centroid

Harte, Sandler and Gasser [29] map a 12-bin chroma vector c (energy per pitch class) to a point in $\mathbb{R}^6$:

$$\zeta(c) = \frac{1}{\|c\|_1} \sum_{\ell=0}^{11} c_{\ell} \Phi_{\ell}, \quad \Phi_{\ell} = (r_1 \sin \frac{7\pi\ell}{6}, r_1 \cos \frac{7\pi\ell}{6}, r_2 \sin \frac{3\pi\ell}{2}, r_2 \cos \frac{3\pi\ell}{2}, r_3 \sin \frac{2\pi\ell}{3}, r_3 \cos \frac{2\pi\ell}{3}),$$

with $(r_1, r_2, r_3) = (1, 1, \frac{1}{2})$. The three coordinate pairs place each pitch class on the circle of fifths, a circle of minor thirds (4 points) and a circle of major thirds (3 points), which are exactly the three directions of the Tonnetz; wrapping the lattice onto these circles is a continuous version of Theorem 3.2. The Euclidean distance between successive centroids is the *harmonic change detection function* used to locate chord changes. Our implementation `tonal_centroid` agrees with `librosa.feature.tonnetz` [36] to within 3×10^{-8} on synthesised audio. In centroid space the closest pairs of triads are *R*-related, followed by *P*- and *L*-related pairs (Figure 8.2).

Python sheet

book/07_corpus_analysis.ipynb reruns Table 8.2 after `bash scripts/fetch_data.sh`; book/08_audio_centroid.ipynb synthesises a progression, computes chroma, tonal centroid and HCDF, and compares with *librosa*.

Questions and answers

- Q.** Why reduce C7 to C before measuring Tonnetz distance?
A. The classical Tonnetz has only major and minor triads as faces; the reduction keeps the triad that carries the chord's root and quality and discards the seventh.
- Q.** What does the within-piece shuffle control for?
A. The chord vocabulary. A piece in one key uses nearby triads anyway; comparing with its own shuffles asks whether the *order* of chords is more local than the vocabulary alone implies.
- Q.** Why is motion by a perfect fifth at distance 2 in Γ ?
A. $C \xrightarrow{L} Em \xrightarrow{R} G$: the two triads share one note (G) and are linked through a triad that shares two notes with each.
- Q.** Name two reasons why pop corpora show more distance-4 transitions.
A. Stepwise root motion between major triads (IV–V, $\flat$ VII–I) is idiomatic in rock and pop, and parallel voice leading is common there; both are “far” in a common-tone metric.
- Q.** How is the tonal centroid related to the Tonnetz?
A. It projects pitch classes onto circles of fifths, minor thirds and major thirds, the three lattice directions of the Tonnetz, and averages over the chord's notes.
- Q.** Suggest a way to test whether the Beethoven result is specific to Beethoven.
A. Run the same pipeline on the Distant Listening Corpus, which uses the same DCML format, grouped by composer and era, and compare observed-minus-shuffled distances.

Software ecosystem and open problems

9.1 Tools

Tool	What it does	Where
TonnetzViz [15]	real-time Tonnetz display of MIDI or keyboard input in the browser	github.com/cifkao/tonnetz-viz
HexaChord [6]	chord complexes, trajectories, compliance measures	Bigo's software page (Java)
music21 [21]	score parsing, chordify, Roman numerals, corpus	<code>pip install music21</code>
librosa [36]	chroma and tonal-centroid (tonnetz) features	<code>pip install librosa</code>
mir_eval [38]	chord-label parsing and evaluation	<code>pip install mir_eval</code>
ChoCo tools [22]	JAMS conversion and harmony knowledge graphs	github.com/smashub/choco
tonnetzkit (this report)	triads, <i>PLR</i> , dual graph, Hamiltonian cycles, chord complexes, homology, edge gluings, corpus loaders	companion repository

9.2 Open problems

The following projects range from student exercises to research questions. Each has a starting point in `book/09_research_sandbox.ipynb`.

1. Canonical edge Tonnetze. Chapter 7 showed that 12 whole-tone shell chords admit about 3×10^{12} edge gluings producing surfaces from the sphere to seven cross-caps. Classify them exactly up to the symmetries of the whole-tone scale, and find a musically motivated gluing rule (for example, requiring adjacent chords to share two notes) that selects a unique surface. Which rules produce the projective plane of the slide?

2. Hamiltonian cycles for seventh chords. The graph on the 48 dominant, minor, half-diminished and major seventh chords, joined when they share three notes, generalises Γ . How many Hamiltonian cycles does it have, and how do they fall into symmetry classes? Exhaustive backtracking needs pruning here; compare the seventh-chord groups of [11] and the generalized dual of [10].

3. Perception. Test whether listeners prefer Hamiltonian cycles to matched controls (Chapter 5), and whether ratings track voice-leading cost (28 versus 36 semitones in Table 5.1).

4. Scaling the corpus study. Apply the pipeline of Chapter 8 to ChoCo and Chordonomicon, stratified by genre and decade. Does the observed-minus-shuffled distance separate styles? Does it change over time?

5. Better metrics. Combine the Tonnetz metric with voice-leading and functional information, or learn a metric from data, and compare with Tonnetz-trajectory descriptors [33] and image-like Tonnetz inputs to neural networks [14].

6. Topological fingerprints. Compute persistent homology of the chord complexes visited by pieces [3], and relate Betti-number signatures to style.

7. Configurations. Boland and Hughston interpret the Tonnetz via its Levi graph as a (12_3) configuration and build analogous tone networks for pentatonic and diatonic systems [7, 8]. Which of their networks carry Hamiltonian cycles, and what are their dual surfaces?

9.3 Conclusion

Euler’s small lattice of fifths and thirds turns out to be a triangulated torus, a Cayley-like graph of a dihedral group, and a coordinate system for real music. Its dual graph has exactly 62 Hamiltonian cycles in eight symmetry classes; its generalizations produce cylinders, spheres, Möbius bands and, once chords are glued along notes, projective planes and higher non-orientable surfaces. Placed against data, it separates the common-tone logic of classical progressions from the whole-step logic of pop. The Tonnetz is best seen not as the one true map of harmony but as a precise mathematical model whose predictions can be computed, tested and refined, which is what the accompanying Python sheets are designed to let readers do.

Questions and answers

- Q. Why is exhaustive search harder for seventh chords than for triads?
A. The graph is larger (48 vertices) and has higher degree, so the number of partial paths grows much faster; pruning (e.g. forcing edges at degree-2 vertices of the remaining graph) or dedicated solvers become necessary.
- Q. What would a positive result in open problem 4 look like?
A. A stable, significant difference in observed-minus-shuffled Tonnetz distance between genres or eras across large corpora, robust to the choice of chord reduction.
- Q. Give one limitation of the Tonnetz as a model of harmonic distance.
A. It ignores function and register and treats stepwise parallel motion between major triads as far, although such motion is idiomatic and not perceived as remote in pop.
- Q. Which chapter’s result would you extend first, and why?
A. Any answer with a reason is acceptable; for example, Chapter 7, because the space of edge gluings is large, little studied and computationally accessible with the provided code.

Repository guide and reproduction

A.1 Layout

```

tonnetz-tutorial/
  tonnetzkit/    core.py (triads, P/L/R)
                  graphs.py (dual graph, Hamiltonian cycles)
                  topology.py (complexes, homology, orientability)
                  edge_tonnetz.py (edge gluings)
                  chords.py (parsers, loaders, tonal centroid)
  book/          00_welcome ... 09_research_sandbox.ipynb, _toc.yml
  scripts/       fetch_data.sh, corpus_analysis.py, make_figures.py
  report/        main.tex, ch*.tex, references.bib, figures/
  results/       corpus_stats.json
  tests/         test_tonnetzkit.py (pins every number in the report)

```

A.2 Reproducing everything

```

pip install -e ".[dev,audio,corpora]"      # or: pip install -r requirements.txt
pytest -q                                  # 8 tests, about 2 s
bash scripts/fetch_data.sh                 # Beethoven (ABC) + NYU MARL chord labels
python scripts/corpus_analysis.py          # Table 8.2 -> results/corpus_stats.json
python scripts/make_figures.py            # all figures -> report/figures/
jupyter-book build book/                  # the Python sheets as a website
cd report && latexmk -pdf main.tex         # this report

```

The GitHub Actions workflow in `.github/workflows/ci.yml` runs the tests on every push and publishes the Jupyter Book to GitHub Pages from the main branch.

A.3 Map from chapters to sheets

Chapter	Sheet	Main functions
2 Preliminaries	01_pitch_classes	<code>pc</code> , <code>transpose</code> , <code>invert</code> , <code>trichord_classes</code>
3 Tonnetz	02_tonnetz_torus	<code>generalized_tonnetz</code> , <code>classify_surface</code>
4 <i>PLR</i>	03_plr_dual	<code>P</code> , <code>L</code> , <code>R</code> , <code>plr_graph</code> , <code>apply_word</code>
5 Hamiltonian	04_hamiltonian	<code>hamiltonian_cycles</code> , <code>symmetry_classes</code>
6 Topology	05_generalized_tonnetze	<code>SimplicialComplex.betti</code> , <code>is_orientable</code>
7 Non-orientable	06_edge_tonnetz	<code>essential_notes</code> , <code>sample_edge_tonnetze</code>
8 Data	07_corpus_analysis, 08_audio_centroid	<code>harte_to_triad</code> , <code>load_dcml_corpus</code> , <code>tonal_centroid</code>
9 Outlook	09_research_sandbox	—

A.4 Data licences

The repository contains no third-party data. `fetch_data.sh` downloads the Annotated Beethoven Corpus (CC BY-NC-SA 4.0; cite [37]) and the NYU MARL chord annotations (cite the repository and the RWC database [27]); Bach chorales are loaded from the music21 corpus. Users must respect each source's terms.

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Bibliography

- [1] G. Albinì and S. Antonini. Hamiltonian cycles in the topological dual of the Tonnetz. In *Mathematics and Computation in Music (MCM 2009)*, volume 38 of *Communications in Computer and Information Science*, pages 1–10. Springer, 2009. doi: 10.1007/978-3-642-02394-1_1.
- [2] G. Albinì and M. P. Bernardi. Hamiltonian graphs as harmonic tools. In *Mathematics and Computation in Music (MCM 2017)*, volume 10527 of *LNCS*. Springer, 2017.
- [3] M. G. Bergomi, A. Baratè, and B. Di Fabio. Towards a topological fingerprint of music. In *Computational Topology in Image Context (CTIC 2016)*, volume 9667 of *LNCS*. Springer, 2016.
- [4] N. Biamonte. Triadic modal and pentatonic patterns in rock music. *Music Theory Spectrum*, 32(2):95–110, 2010.
- [5] L. Bigo and M. Andreatta. Topological structures in computer-aided music analysis. In D. Meredith, editor, *Computational Music Analysis*. Springer, 2016.
- [6] L. Bigo, J.-L. Giavitto, M. Andreatta, O. Michel, and A. Spicher. Computation and visualization of musical structures in chord-based simplicial complexes. In *Mathematics and Computation in Music (MCM 2013)*, volume 7937 of *LNCS*, pages 38–51. Springer, 2013. doi: 10.1007/978-3-642-39357-0_3.
- [7] J. R. Boland and L. P. Hughston. Configurations, tessellations and tone networks. arXiv:2505.08752, 2025.
- [8] J. R. Boland and L. P. Hughston. Tonnetz theory, classical harmony, and the combinatorial geometry of abstract musical resources. arXiv:2604.19960, 2026.
- [9] C. Callender, I. Quinn, and D. Tymoczko. Generalized voice-leading spaces. *Science*, 320(5874):346–348, 2008.
- [10] S. Cannas and M. Andreatta. A generalized dual of the Tonnetz for seventh chords: Mathematical, computational and compositional aspects. In *Proceedings of Bridges 2018: Mathematics, Art, Music, Architecture, Education, Culture*, pages 301–308, 2018.
- [11] S. Cannas, S. Antonini, and L. Pernazza. On the group of transformations of classical types of seventh chords. In *Mathematics and Computation in Music (MCM 2017)*, volume 10527 of *LNCS*, pages 13–25. Springer, 2017.
- [12] G. Capuzzo. Neo-Riemannian theory and the analysis of pop-rock music. *Music Theory Spectrum*, 26(2):177–199, 2004.
- [13] M. J. Catanzaro. Generalized Tonnetze. *Journal of Mathematics and Music*, 5(2):117–139, 2011. doi: 10.1080/17459737.2011.614448.
- [14] C.-H. Chuan and D. Herremans. Modeling temporal tonal relations in polyphonic music through deep networks with a novel image-based representation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.
- [15] O. Cífka. TonnetzViz: an app for visualizing music using the Tonnetz. <https://github.com/cifkao/tonnetz-viz>, 2016.
- [16] R. Cohn. Maximally smooth cycles, hexatonic systems, and the analysis of late-romantic triadic progressions. *Music Analysis*, 15(1):9–40, 1996.

- [17] R. Cohn. Neo-Riemannian operations, parsimonious trichords, and their “Tonnetz” representations. *Journal of Music Theory*, 41(1):1–66, 1997.
- [18] R. Cohn. Introduction to neo-Riemannian theory: A survey and a historical perspective. *Journal of Music Theory*, 42(2):167–180, 1998.
- [19] R. Cohn. *Audacious Euphony: Chromaticism and the Triad’s Second Nature*. Oxford University Press, 2012.
- [20] A. S. Crans, T. M. Fiore, and R. Satyendra. Musical actions of dihedral groups. *American Mathematical Monthly*, 116(6):479–495, 2009.
- [21] M. S. Cuthbert and C. Ariza. music21: A toolkit for computer-aided musicology and symbolic music data. In *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR)*, 2010.
- [22] J. de Berardinis, A. Meroño-Peñuela, A. Poltronieri, and V. Presutti. ChoCo: A chord corpus and a data transformation workflow for musical harmony knowledge graphs. *Scientific Data*, 10:641, 2023. doi: 10.1038/s41597-023-02410-w.
- [23] T. de Clercq and D. Temperley. A corpus analysis of rock harmony. *Popular Music*, 30(1): 47–70, 2011.
- [24] J. Douthett and P. Steinbach. Parsimonious graphs: A study in parsimony, contextual transformations, and modes of limited transposition. *Journal of Music Theory*, 42(2):241–263, 1998.
- [25] L. Euler. *Tentamen novae theoriae musicae ex certissimis harmoniae principiis dilucide expositae*. Typographia Academiae Scientiarum, Saint Petersburg, 1739.
- [26] E. Gollin. Some aspects of three-dimensional Tonnetze. *Journal of Music Theory*, 42(2): 195–206, 1998.
- [27] M. Goto, H. Hashiguchi, T. Nishimura, and R. Oka. RWC music database: Popular, classical and jazz music databases. In *Proceedings of the 3rd International Conference on Music Information Retrieval (ISMIR)*, 2002.
- [28] C. Harte, M. Sandler, S. Abdallah, and E. Gómez. Symbolic representation of musical chords: A proposed syntax for text annotations. In *Proceedings of the 6th International Conference on Music Information Retrieval (ISMIR)*, 2005.
- [29] C. Harte, M. Sandler, and M. Gasser. Detecting harmonic change in musical audio. In *Proceedings of the 1st ACM Workshop on Audio and Music Computing Multimedia*, pages 21–26, Santa Barbara, CA, 2006. doi: 10.1145/1178723.1178727.
- [30] J. Hentschel, Y. Rammos, M. Neuwirth, and M. Rohrmeier. A corpus and a modular infrastructure for the empirical study of (an)notated music. *Scientific Data*, 12:671, 2025.
- [31] F. D. Jevtić and R. T. Živaljević. Generalized Tonnetz and discrete Abel–Jacobi map. *Topological Methods in Nonlinear Analysis* (2021); arXiv:2002.09184, 2021.
- [32] S. Kantarelis, K. Thomas, V. Lyberatos, E. Dervakos, and G. Stamou. CHORDONOMICON: A dataset of 666,000 songs and their chord progressions. arXiv:2410.22046, 2024.
- [33] E. Karystinaios, C. Guichaoua, M. Andreatta, L. Bigo, and I. Bloch. Musical genre descriptor for classification based on Tonnetz trajectories. In *Journées d’Informatique Musicale (JIM)*, Strasbourg, 2020.

- [34] D. Lewin. *Generalized Musical Intervals and Transformations*. Yale University Press, New Haven, 1987.
- [35] R. Lieck, F. C. Moss, and M. Rohrmeier. The tonal diffusion model. *Transactions of the International Society for Music Information Retrieval*, 3(1):153–164, 2020.
- [36] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto. librosa: Audio and music signal analysis in Python. In *Proceedings of the 14th Python in Science Conference*, 2015.
- [37] M. Neuwirth, D. Harasim, F. C. Moss, and M. Rohrmeier. The annotated Beethoven corpus (ABC): A dataset of harmonic analyses of all Beethoven string quartets. *Frontiers in Digital Humanities*, 5:16, 2018. doi: 10.3389/fdigh.2018.00016.
- [38] C. Raffel, B. McFee, E. J. Humphrey, J. Salamon, O. Nieto, D. Liang, and D. P. W. Ellis. mir_eval: A transparent implementation of common MIR metrics. In *Proceedings of the 15th International Society for Music Information Retrieval Conference (ISMIR)*, 2014.
- [39] H. Riemann. *Vereinfachte Harmonielehre, oder die Lehre von den tonalen Funktionen der Akkorde*. Augener, London, 1893.
- [40] D. Tymoczko. *A Geometry of Music: Harmony and Counterpoint in the Extended Common Practice*. Oxford University Press, 2011.
- [41] D. Tymoczko. The generalized Tonnetz. *Journal of Music Theory*, 56(1):1–52, 2012.
- [42] A. von Oettingen. *Harmoniesystem in dualer Entwicklung: Studien zur Theorie der Musik*. W. Gläser, Dorpat and Leipzig, 1866.
- [43] W. Xu, E. Hall, and M. Rohrmeier. Diachronic modeling of tonal coherence on the Tonnetz across classical and popular repertoires. arXiv:2603.27035, 2026.
- [44] J. Yust. Generalized Tonnetze and Zeitnetze, and the topology of music concepts. *Journal of Mathematics and Music*, 14(2), 2020. doi: 10.1080/17459737.2020.1725667.